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Counteracting effects of "hook avoidance" and "hook habituation" on angler catch rates in a catch-and-release fishery

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Abstract

Catch-and-release (C&R) angling is often used to maintain high catch rates but fish vulnerability to capture may decrease following hooking, thereby decreasing angler catch per unit effort (CPUE) (hyperdepletion). To determine if fish post-capture response affected recapture probability and population-level CPUE, individual capture histories of Largemouth Bass in two lakes were compared before and after doubling angling effort in a Before-After Control-Impact (BACI) analysis. Previous capture and day-of-season both affected recapture probability. Counteracting effects of previous capture and reduced late-season catch rates caused no hyperdepletion of angler CPUE. Our results highlight the complexity of fish behavioral responses to angling and suggest that hyperdepletion of angling catch rates may not be an issue in C&R fisheries.

KEYWORDS
catch and release, catch rates, hook learning, hyperdepletion, largemouth bass, mark–recapture model

1 | INTRODUCTION

Because catch rates are often directly tied to angler satisfaction (Hilborn & Walters, 1992; Ward et al., 2013), many recreational fisheries have shifted to catch-and-release regulations or anglers have voluntarily adopted catch-and-release (Sass & Shaw, 2020, Cooke & Suski, 2005, Barnhart, 1989). Catch-and-release fisheries often aim to maximize catch rates by removing harvest to maintain high population abundance, although catch-and-release can reduce catch rates (Camp et al., 2015; Sass & Shaw, 2020). If angling encounters influence fish behavior that reduces catchability, catch-and-release angling could reduce catch rates, a form of hyperdepletion where catch rates decline without a change in fish abundance (Askey et al., 2006; Camp et al., 2015; Kuparinen et al., 2010).

Although hyperdepletion of angler catch rates could theoretically be caused by individual fish learning to avoid fishing gear (hook learning) or other post-release behavioral changes, most studies have focused on either population-scale response of angler catch per unit effort (CPUE) to effort (hyperdepletion) (Alós et al., 2015; Pierce & Tomcko, 2003; Wegener et al., 2018) or individual-scale post-release behavior, including hook learning (Klefoth et al., 2013; Louison et al., 2018; Wallerius et al., 2020). For example, individual capture histories from experimental angling of Rainbow Trout (*Oncorhynchus mykiss*) were negatively affected by the number of times a fish was previously caught (referred to as learned hook avoidance), intra-population heterogeneity in catchability, and a seasonal decrease in catchability that was independent of angling effort (Askey et al., 2006). Understanding how previous capture and resulting fish behavior influence angler catch rates is especially important for managing catch-and-release fisheries that cannot rely on traditional management practices, such as bag and size limits, to increase CPUE, but instead may need to modify angler effort by spatial or temporal fishery closures (van Poorten & Post, 2005).

Increased effort frequently depletes CPUE, although the magnitude of reduced CPUE attributable to angler effort varies among

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species. For example, consistent and increased angler effort decreased CPUE at constant abundances of Rainbow Trout, Brown Trout (*Salmo trutta*), Common Carp (*Cyprinus carpio*), Largemouth Bass (*Micropterus nigricans*), and Pike (*Esox lucius*) (Askey et al., 2006; Beukema, 1969, 1970; Kuparinen et al., 2010; Raat, 1985; Sass et al., 2018; van Poorten & Post, 2005; Wegener et al., 2018; Young & Hayes, 2004). Decreased angling effort can allow CPUE to rebound or delay declines in CPUE in Rainbow Trout (Koeck et al., 2019), Largemouth Bass (Wegener et al., 2018), and simulated lake fisheries (Camp et al., 2015). Such consistent patterns of hyperdepletion are most often attributed to fish entering a post-release recalcitrant state that could reflect fish changing habitat use, temporarily ceasing foraging, or learning to avoid hooks (Camp et al., 2015; Cox, 2000; Cox & Walters, 2002), although individual, post-release behavior is not often measured.

A change in behavior following capture that leads to reduced probability of subsequent capture could be attributable to a change in habitat use, foraging behavior, search image, or learning to avoid lures. Growth temporarily declines following hooking, consistent with reduced foraging after hooking and capture (Cline et al., 2012). Individual capture histories also indicate that fish are less likely to be captured again after capture (Askey et al., 2006). In laboratory and pond studies, fish can learn how to avoid hooks (Hessenauer et al., 2016; Wallerius et al., 2020). Intra-population heterogeneity in behavior (bold vs. shy syndromes) may also influence vulnerability to angling and angler catch rates, either directly or through hook-avoidance learning (Frost et al., 2007; Hackney & Linkous, 1978; Louison et al., 2018; Philipp et al., 2009; Wilson et al., 2015; Wilson & Godin, 2009).

Although catch-and-release could interact with post-release behavioral changes to cause catch rate hyperdepletion, many non-behavioral factors also influence angler catch rates. For example, complex interactions between fish internal state and fishing gear characteristics affect capture vulnerability (Lennox et al., 2017). Furthermore, angler catch rates are influenced by ecological factors, such as seasonality. Within-season decreases in CPUE are caused by changes in resource availability, water temperature, or temporal changes in fish nesting or aggression behavior (Kuparinen et al., 2010; van Poorten & Post, 2005). Other non-behavioral factors thought to influence catch rates include recruitment of individuals to angler gear, angler skill, and effort sorting (Lennox et al., 2017; van Poorten et al., 2016; Ward et al., 2013). If the effect of non-behavioral factors, such as seasonality, are large enough, anglers may be unlikely to detect hyperdepletion.

Our objective was to test if previous capture altered the probability of subsequent capture in a popular catch-and-release angling species (Largemouth Bass), and if so, if effects of previous capture caused detectable hyperdepletion of angler catch rates. Based on individual capture histories and CPUE from daily experimental angling in two private lakes, we fit a recapture probability model to quantify effects of previous capture and other potential predictors of recapture probability. We then used changes in angler catch rate in a Before-After Control-Impact (BACI) manipulation of effort, with output from a simulation model informed by the statistical recapture probability model, to determine if post-release declines in vulnerability to angling could drive detectable hyperdepletion within a recreational fishery.

2 | METHODS

2.1 | Data collection

The study was conducted on Peter (2.5 ha) and Paul (1.7 ha) Lakes at the University of Notre Dame Environmental Research Center in Gogebic County Michigan (46° 15′ 10.3674, "−89° 30′ 12.924"). These morphometrically similar, private lakes have fish communities with Largemouth Bass, Bluegill (*Lepomis macrochirus*), and Pumpkinseed (*Lepomis gibbosus*), and have no public fishing access. The lakes have a long history of whole-lake experiments, but fish communities have not been manipulated in recent years (Carpenter, 1987; Carpenter et al., 2022).

Standardized rod-and-reel angling was used to catch Largemouth Bass in both lakes 5 days per week from June 1, 2022, to August 5, 2022. Two anglers fished each basin for 2 h per day (increased to 4 h per day during effort manipulation) using only 5″ soft-plastic YUM Dinger stick baits (varied colors) and barbless size-3 hooks. Anglers fished the whole shoreline on each sampling date. The same two anglers fished both lakes for the entire week, alternating the lake fished in the morning and afternoon each day, but anglers were rotated each week from a pool of four anglers. Fish that were captured by angling were placed into a 95-liter live well with a bubbler to maintain oxygen levels during recovery. Within an hour of capture, fish were checked for an existing PIT tag, weighed (g), measured in total length (mm), and tagged, if not previously tagged, with a 12 mm PIT tag (Oregon RFID; Portland, OR). Fish were not harvested, but six fish were removed from Peter Lake and one fish was removed from Paul Lake because of hooking mortality.

Alternating-current boat electrofishing was used in early summer (from May 27 to June 5, 2022) and again in the fall (August 15, 2022) in both lake basins to measure changes in relative abundance over the study period. Each electrofishing trip covered the entire shoreline with 2 netters and a driver along a saw-tooth pattern. Electrofishing was also used to estimate a minimum number of fish that were never captured by angling. Fish captured were measured and marked using the same methods as fish caught by angling.

We expected catch rates to decline as effort increased in the absence of a change in abundance, if hyperdepletion was evident. To test for hyperdepletion, angling effort was doubled in Peter Lake midway through the sampling period in a BACI design. Paired daily angling in each lake between June 1 and July 4 was the reference period for both lakes at four angler-hours of effort each day. On July 5, 2022, the angling effort in the treatment lake (Peter) was doubled to eight angler-hours each day.

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2.2 | Data analysis

To test for effects of previous capture on probability of recapture of individual fish by angling, Cormack–Jolly–Seber (CJS) mark–recapture models were fit using the package "marked" in R (Laake et al., 2013; Team, 2023). A suite of candidate, mixed-effect CJS models were fit with 2 hyperparameters: ϕ_{ij} was the probability that fish i survived between sampling occasions j and $j+1$, and p_{ij} was the probability that fish i was captured on sampling occasion j. Hyperparameters estimated the probability of animal i's capture history, Prob(CH_i), given its first capture (Pledger et al., 2003). Survival was constant across time and among individuals ϕ_{ij} (~1) and recapture probability p_{ij} was determined by factors that differed across time or among individuals. Two types of factors that would affect p_{ij} were considered: (1) effects of previous capture; and (2) covariates for individual variation among fish, fish size, and day of year (Equation 1; Table S1). Model performance was assessed using AIC and whether Hessian-based 95% confidence intervals of coefficients included zero (Laake et al., 2013). The sign and magnitude of model coefficients were interpreted for biological meaning.

$$\mathrm{Prob}(\mathrm{CH}_i) = \phi_{ij}\,(\sim 1) \times \mathrm{p}_{ij}\,(\sim \text{previous capture effects} + \text{covariates}) \quad (1)$$

To test the effect of previous capture, two variables and their interaction were used that were previously identified as important (Equation 1; Askey et al., 2006, Koeck et al., 2019). Longer term cumulative effects of multiple captures during the season were tested using the total number of captures prior to time t (c); and an acute effect of the most recent capture as days since the last encounter at time t (d). Factorial combinations of these metrics were considered for the capture portion of the probability of capture (p_{ij}) for fish i at time t in our CJS model. Four candidate models were:

$$\mathrm{Prob}(\mathrm{CH}_i) = \phi_{ij}\,(\sim 1) \times \mathrm{p}_{ij}\,(\sim \mathrm{d} + \text{covariates}) \quad (2)$$

$$\mathrm{Prob}(\mathrm{CH}_i) = \phi_{ij}\,(\sim 1) \times \mathrm{p}_{ij}\,(\sim \mathrm{c} + \text{covariates}) \quad (3)$$

$$\mathrm{Prob}(\mathrm{CH}_i) = \phi_{ij}\,(\sim 1) \times \mathrm{p}_{ij}\,(\sim \mathrm{d} + \mathrm{c} + \text{covariates}) \quad (4)$$

$$\mathrm{Prob}(\mathrm{CH}_i) = \phi_{ij}\,(\sim 1) \times \mathrm{p}_{ij}\,(\sim \mathrm{d} + \mathrm{c} + \mathrm{d}:\mathrm{c} + \text{covariates}) \quad (5)$$

Covariates were added to candidate previous-capture models ($N=4$; Equation 2–5) that were previously found to influence individual capture history (Askey et al., 2006). Covariates included fish size (length at first capture, mm), intra-season environmental variation (day of year, *doy*), and individual fish variation (fish ID). Size at first capture and fish ID were time-invariant covariates, with fish ID as a random effect. Day of year was a time-varying covariate. The total number of models ($N=32$) included eight combinations of covariates, including a model with no covariates, and four candidate post-release-behavior models (Table S1), all of which used individual capture histories for 224 fish. To quantify effect sizes of factors in the best performing model, the model was refit with z-transformed covariates.

Abundance of fish susceptible to angling (>133 mm, the minimum length of a fish captured during angling) was estimated in each lake using the Schnabel (1938) estimator for mark–recapture angling data. Lake-specific CPUE was estimated using daily captures each day, divided by angling effort (time spent angling multiplied by the number of anglers). To test for an effect of increased angler effort on CPUE (fish captured per angling hour) from the BACI-design, Welch's t-test was used to compare the differences in daily CPUE between lakes before and after angler effort was doubled (Stewart-Oaten et al., 1992).

2.3 | Capture history simulation

To interpret how previous capture and covariate effects from the statistical, individual recapture model (Equation 1) interacted to influence seasonal dynamics of angler CPUE, capture histories were simulated. Simulations with a full model represented mean expectations for seasonal angler CPUE dynamics, and the influence of individual factors were assessed by removal of each factor (e.g., days since last capture or day of year). In simulations, individual fish were randomly sampled, weighted by individual preceding simulated capture history, to determine whether each fish in the simulated fish population was captured or not. A given simulation was defined by the number of fish in the population (N), the day of year when the fishing season began, the number of days in a fishing season (N_{days}), the proportion of a population that saw a bait for each hour of effort (q'), and hours of angling effort each day (E_t).

Each simulation treated modeled fish as naïve to capture (days since capture was set to 75 and their cumulative capture was 0). The probability that a modeled fish was presented with a bait was independent of simulated capture history. Therefore, on day t, $E_t{*}N{*}q'$ fish were randomly selected to see a bait. Population size ($N=100$), season length (57 days) and effort ($E_t=2$) were similar to experimental angling. The value for q' (0.025) was set to generate early season angler CPUE similar to observed experimental angling. The probability of capture for a fish that saw a bait was related to preceding simulated capture history. For each day of the simulation, the presence or absence of capture was recorded for each fish in a capture history matrix. Once a simulation was complete, catch per unit effort (CPUE) was computed for the simulated population based on the set of simulated capture histories.

Five scenarios used subsets of effects from the best-performing, statistical model, to visualize individual effects of intra-season environmental factors (*doy*) and different components of previous capture on dynamics of angler CPUE (d, c, and $d{:}c$, Table 1). Simulations were probabilistic and simulated CPUE varied among iterations, so 500 replicate simulations were used to estimate mean CPUE (±95% confidence intervals) for each scenario. Finally, detectability of hyperdepletion in field manipulations of effort was evaluated with two sets of 500 simulations and the full p(capture) model, where angler effort was constant at four angler-hours per day across the fishing season (reference) or doubled from four to eight angler-hours

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TABLE 1 Simulation scenarios to evaluate the probability of recapture (p(capture)) in relation to days since capture (*d* = hook avoidance), cumulative captures (*c* = hook habituation), their interaction (*d:c* = temporal change in hook avoidance), and day of year (*doy* = environmental variation) for standardized rod-and-reel angling for largemouth bass in Peter and Paul lakes at the University of Notre Dame Environmental Research Center in Gogebic County Michigan from June 1, 2022, to August 5, 2022.

Scenario	Equation	Description
1	p(capture) = d + c + d: c + doy	Intra-season + previous-capture
2	p(capture) = doy	No previous capture
3	p(capture) = d + c + d: c	No intra-season
4	p(capture) = d	Days since capture
5	p(capture) = d + doy	Days since capture + intra-season

halfway through the season (treatment). Welch's t-tests were used to test for an effect of effort doubling on simulated CPUE for a single simulation and for the average of 500 simulations.

3 | RESULTS

Experimental angling caught 114 unique fish 295 times in Peter Lake and 111 unique fish 311 times in Paul Lake. Individual fish ranged from 0 to 9 in the number of angling captures (Figure 1), but 13% of individual fish in Peter and 15% in Paul were captured five times or more within the sampling season. For individual fish caught at least twice, the number of days between angling capture events ranged from 1 to 53 days. Abundance of largemouth bass in Peter Lake was 121 fish (95% confidence interval, 106–142) and in Paul Lake was 120 fish (95% CI, 106–139).

Density of fish (electrofishing CPUE) that could be captured by anglers (>133 mm) did not decline across the experiment in either lake. In Paul Lake, electrofishing CPUE (9.3 fish per hour) on August 15, 2022, was within the range of electrofishing CPUE at the beginning of the experiment (6.9–11.2 fish per hour; *N* = 2). In Peter Lake, electrofishing CPUE (9.6 fish per hour) on August 15, 2022, was above the range of electrofishing CPUE at the beginning of the experiment (1–7.4 fish per hour; *N* = 3). Of 20 fish longer than 133 mm caught by electrofishing, 20 (8%) were never captured by anglers. Half of the 20 fish that were captured by electrofishing, but never caught by angling, were shorter than 173 mm at the end of the study.

3.1 | Previous capture effects

Previous capture affected subsequent recapture probability. The best previous-capture model included days since last capture (*d*), cumulative capture (*c*), and the interaction between these covariates (*d:c*) (Table S1). The probability of recapture increased with time since last capture (*d*) and with subsequent captures (*c*). The rate at

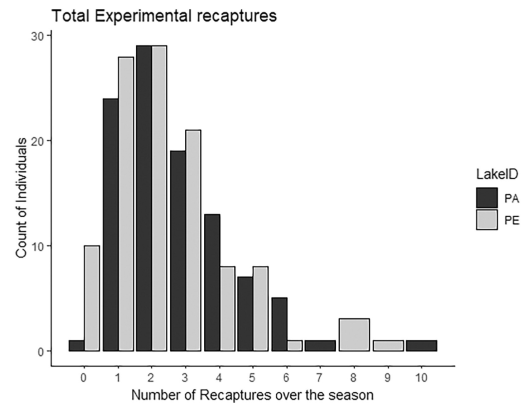

FIGURE 1 Frequency of captures of individual largemouth bass by standardized rod-and-reel angling for largemouth bass in Peter (*N* = 114) and Paul (*N* = 111) lakes at the University of Notre Dame Environmental Research Center in Gogebic County Michigan from June 1, 2022, to August 5, 2022. Recapture frequencies of zero reflect fish captured only during electrofishing and never captured via angling that were in the population and large enough to be susceptible to angling (≥133 mm).

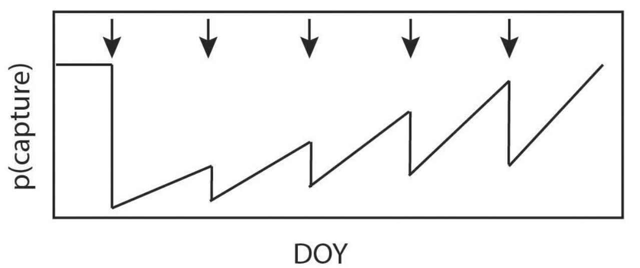

FIGURE 2 Conceptual model of previous-capture effects (ignoring intra-season environmental effects) on recapture probability p(capture) over days of the year (DOY). Each arrow denotes a capture event for an individual fish. The probability of capture, p(capture), drops immediately following each capture, but increases over time due to a positive effect of days since capture (*d* = hook avoidance). The p(capture) following each subsequent capture increases due to a positive effect of cumulative capture (*c* = hook habituation). A positive interaction term between *d* and *c* (temporal change in hook avoidance) also increases the rate of recovery of p(capture) for later capture events. Although time between capture events should be influenced by p(capture), time between capture events was held constant to visualize the three component effects on capture history.

which capture probability increased following a capture should increase with subsequent captures (Figure 2).

Capture probability declined within the angling season (*doy*), but individual average capture probability did not vary significantly among individuals (fish ID). Covariate models that did not include a day-of-year effect were ~300 AIC points worse than covariate models that did (Table 2). The covariate model with day-of-year alone had the lowest AIC, but models that included fish length or a random, individual fish effect were within 2–12 AIC points of the day of year covariate model. However, 95% confidence intervals for additional parameters included zero (Equation 6).

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TABLE 2 Candidate models for evaluating the probability of recapture (p(capture)) in relation to day of year (doy = environmental variation), body size (size), and individual fish (fish ID = intra-population heterogeneity) for standardized rod-and-reel angling for largemouth bass in Peter and Paul lakes at the University of Notre Dame Environmental Research Center in Gogebic County Michigan from June 1, 2022, to August 5, 2022.

Model	AIC	Delta AIC	Covariates with 0s in CIs
Doy	2359.8	0	–
Size + doy	2361.5	1.7	Size
Fish ID + doy	2368.3	8.5	–
Size + doy + fish ID	2370.1	10.3	Size
None	2628.4	268.6	–
Size	2630.3	270.5	Size
Fish ID	2630.4	270.6	Fish ID
Size + fish ID	2632.3	272.5	Fish ID, size

Note: Fish ID was treated as a random effect in the model and all other effects were fixed effects. Results reported are for the indicated candidate covariate model included in the best performing post-release-behavior model (d + c + d:c; hook avoidance + hook habituation + temporal change in hook avoidance). Covariate estimates that included 0 within 95% confidence intervals are indicated. The best performing model AIC is used to calculate Delta AIC for all other models (d + c + d:c + doy).

$$\text{Prob}(\text{CH}i) = \phi ij\,(\sim 1) \times pij\,(\sim d + c + d{:}\,c + \text{doy}) \tag{6}$$

Effect sizes of all covariates in the best-performing model were within a factor of 2.5 of each other (Table S2).

3.2 | Catch per unit effort effects

Average dynamics of the best model indicated that CPUE was strongly driven by intra-season environmental effects (*doy*), especially late in the season (Figure 3, black line). However, previous-capture effects affected CPUE early in the season (Figure 3). Days since capture (*d*) positively affected recapture probability, with recapture probability staying low until time passed. The positive cumulative capture (*c*) effect counteracted the effect of days since capture (*d*) on angler CPUE by the end of the season. The counter-acting effect of cumulative capture was apparent when the intra-season effect was present or absent. Both intra-season and previous-capture effects on CPUE were less apparent in a single model simulation (95% confidence intervals for all scenarios overlapped) because the angling process was noisy (Figure S1).

Experimental doubling of angler effort did not result in a detectable difference in CPUE between control and treatment lakes. Daily CPUE ranged 0.12–3.75 in the treatment lake (mean = 1.36; Peter Lake) and 0.25–4.29 in the reference lake (mean = 1.66; Paul Lake). In both lakes, CPUE declined over the sampling season

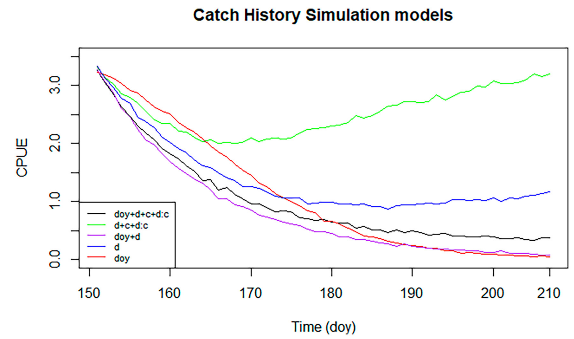

FIGURE 3 Simulated CPUE using different versions of an angling recapture probability model in relation to days since capture (*d* = hook avoidance), cumulative captures (*c* = hook habituation), the interaction (*d:c* = temporal change in hook avoidance), and day of year (*doy* = environmental variation) based on standardized rod-and-reel angling for largemouth bass in Peter and Paul lakes at the University of Notre Dame Environmental Research Center in Gogebic County Michigan from June 1, 2022, to August 5, 2022. The black line reflects the best-performing recapture history model that included hook avoidance, hook habituation, temporal change in hook avoidance, and environmental variation. The green line removed the environmental variation but retained all hook avoidance and habituation components of the model. The blue line included only hook avoidance, the red line included only environmental variation, and the purple line included environmental variation and hook avoidance. Lines are daily average observed angler CPUE across 500 simulations; 95% confidence intervals of CPUE from these simulations are presented in Figure S1.

(June–August, Figure 4a). Doubling of angling effort had no significant effect on angler CPUE (Welch $t = -0.30$; $df = 35.29$; $p = 0.77$). Simulations of individual capture histories with a mid-season doubling of angling effort were similar to results in Peter and Paul lakes (Welch $t = 0.78$, $df = 54.45$, p-value = 0.44, Figure 4b). Average simulated angler CPUE from 500 iterations of the simulation model indicated that doubling the angler effort significantly reduced angler CPUE, but increased effort only reduced angler CPUE by ~0.1 fish per angler hour (Welch $t = 7.38$, $df = 53.52$, p-value = 1.04e-09, Figure 4c).

4 | DISCUSSION

We found that prior capture reduced the probability a fish would be caught again by an angler, although the probability of recapture increased with time and with the total number of previous captures; like previous studies, a specific mechanism cannot be ascribed (Askey et al., 2006; Koeck et al., 2019; Wegener et al., 2018). Hook avoidance does not appear to be permanent, but rather the probability of recapture appears to gradually increase, consistent with the vulnerable pool dynamics model (Camp et al., 2015). For example, according to our statistical capture history model, the probability of a fish being recaptured increased by 1%–20% per day, depending

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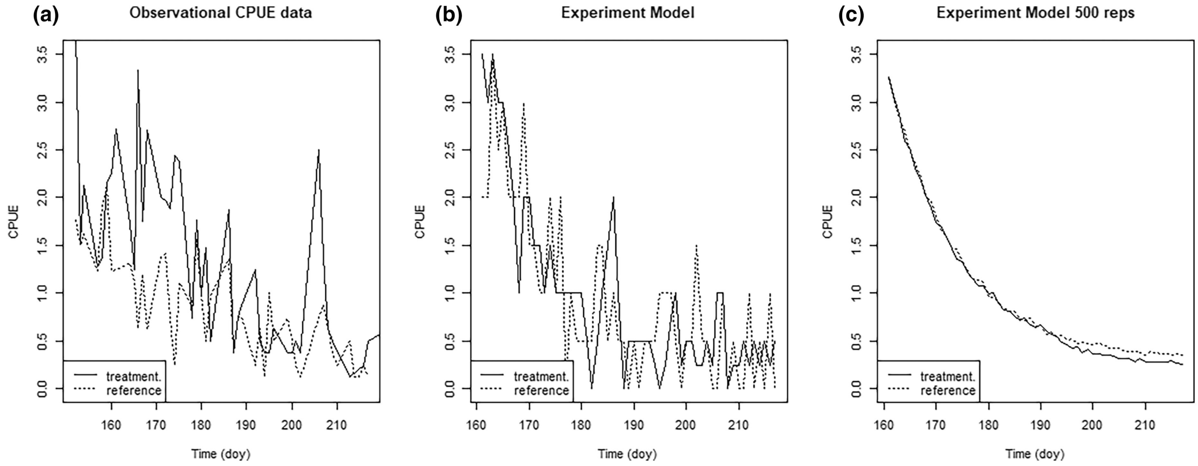

FIGURE 4 Angler CPUE of largemouth bass in relation to day of year in treatment (Peter Lake) and reference (Paul Lake) lakes where effort was doubled from 4 standardized angler-hours of effort per day on days 150–180 to 8 standardized angler-hours per day on day 180 in the treatment lake at the University of Notre Dame Environmental Research Center in Gogebic County Michigan from June 1, 2022, to August 5, 2022 (Panel a). Catch-history simulations were derived from a statistical recapture probability model were used to simulate effects of effort manipulation. A single simulation is shown in Panel b and the average of 500 simulations is shown in Panel c.

on capture history and time of year, in contrast to previous studies where more captures reduced the probability of recapture (Askey et al., 2006; Lennox et al., 2016). Interestingly, we found that hook habituation accelerated the rate of increase in recapture, to further reduce the time captured fish spent in a recalcitrant state of reduced vulnerability to angling (Figure 2). Hook habituation time may depend on factors, such as lure type, use of live bait, and extent of handling between capture and release (Folkedal et al., 2018; Lennox et al., 2017).

Temporary hook avoidance following capture that we observed may be a behavioral response to hook learning, a physiological recovery period from handling stress, or a shift in habitat use away from areas fished by anglers (Klefoth et al., 2013). For example, in tank and pond studies, hook-learning may have been driven by a behavioral change (Hessenauer et al., 2016; Klefoth et al., 2013; Koeck et al., 2020; Raat, 1985). Furthermore, in natural populations, fish lost weight or grew slower after hooking, which could indicate use of suboptimal habitat (Cline et al., 2012). Despite our inability to show that hook learning was the mechanism driving hook avoidance, hook habituation seems more likely to be learned and may be a form of hook learning that has the opposite effect of what has been previously demonstrated (Askey et al., 2006; Folkedal et al., 2018; Wallerius et al., 2020).

We did not detect inherent, individual heterogeneity in recapture probability, unlike previous laboratory and field studies that found consistent differences in individual fish behaviors related to foraging, refuge seeking, and angling (Frost et al., 2007; Klefoth et al., 2017; Louison et al., 2019; Wilson & Godin, 2009). Individual fish are often classified along a behavioral axis from bold to shy that varies among species and individuals in a population (Klefoth et al., 2017; Louison et al., 2018). For example, less aggressive individual muskellunge were more likely to be captured (Bieber et al., 2023). In contrast, aggressive or faster learning Largemouth Bass were more vulnerable to capture than those that were less aggressive or slower learning (Louison et al., 2019; Suski & Philipp, 2004). Furthermore, susceptibility to angling is heritable in largemouth bass (Philipp et al., 2009). More research is needed to determine relative contributions of genetics and individual variation in behavior on the likelihood of recapture to angling.

We found no evidence that experimental doubling of angler effort caused hyperdepletion, perhaps because: (1) initial angling effort density was high enough to cause hyperdepletion prior to our effort manipulation; (2) hook habituation counteracted hook avoidance; (3) intra-season environmental variation significantly reduced catch rates in both lakes more than hyperdepletion; or (4) angling is much too inherently noisy to detect hyperdepletion in only two lakes. First, our experimental angling effort density (approximately 2 angler-hours per hectare per day) prior to doubling was high relative to the angling effort density in the Northern Highland Lake District (2×10^{-5}–2×10^{-2} angler-hours per hectare per day) and may have caused hyperdepletion prior to effort to our effort manipulation (Wisconsin Department of Natural Resources, 2023). Second, our simulations suggested that hook habituation alone could ameliorate some hyperdepletion caused by hook avoidance Third, we found no reduction in CPUE caused by reduced capture probability, because intra-season environmental variation strongly influenced recapture probability, especially at the end of summer. Other studies found declining catch rates despite constant catchability of individuals were caused by intra-season environmental effects (Askey et al., 2006; van Poorten & Post, 2005). For example, CPUE of Rainbow Trout in both exploited and unexploited populations declined over summer because intra-season environmental effects in the unexploited population were larger than fish behavior changes in the exploited population (van Poorten & Post, 2005). Similarly, temporal shifts in fish behavior result from reasons other than hook learning or changes in the number of vulnerable individuals, such as changes in temperature

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and resource availability (Martin, 1958; van Poorten & Post, 2005). Fourth, variability in angler skill, technology, and gear selectivity make angling catch rate data too variable to detect other than very large levels of hyperdepletion (Cooke et al., 2021; Detmer et al., 2020; van Poorten et al., 2016).

Our results suggested that for largemouth bass in small, north temperate lakes, fisheries managers may not need to be concerned about impacts of previous capture on angler CPUE, unlike previous work that detected declines in angler CPUE hyperdepletion and implicated post-release behavioral change as a cause (Askey et al., 2006; van Poorten et al., 2016). Given these previous findings, fisheries managers in a world with ever-increasing catch-and-release angling should be concerned about the impacts of hyperdepletion on angler satisfaction and, perhaps should consider controlling angler effort (Camp et al., 2015). In contrast, our findings suggest that fishery managers may not need to take actions to mitigate hyperdepletion within similar recreational fisheries because (1) fish seem to avoid and habituate in response to angling, (2) angler CPUE declines within the angling season in response to environmental variation, (3) angler CPUE is highly variable, and (4) angler effort density is much lower than would cause hyperdepletion of catch rates. Furthermore, lower effort in north-temperate largemouth bass fisheries during winter may reset effects of capture history in the previous season. Alternatively, if habituation accrues across years, previous capture history may not induce hyperdepletion. However, other possible negative outcomes from increased angler effort in catch-and-release fisheries are possible, including size-structure shifts, post-release mortality, or long-term selective pressure (Hessenauer et al., 2018; Kerns et al., 2016; Philipp et al., 2009). More research is needed to guide managers. Future studies should attempt to disentangle avoidance and habituation behaviors in other fisheries and species. In addition, quantifying plastic, capture-history-induced behaviors, behavioral syndromes (e.g., bold vs. shy), and their interactions may allow for a more thorough assessment of the likelihood of evolution impacting catch-and-release fisheries.

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CONFLICT OF INTEREST STATEMENT

There are no conflicts of interest to disclose.

DATA AVAILABILITY STATEMENT

The data that support the findings of this study are openly available in Zenodo at https://zenodo.org/record/8407883.

ETHICS STATEMENT

All animals were handled and collected under University of Notre Dame IACUC protocols and Michigan Scientific collector's permit #FSCP02092022115909.

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